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Discrete Random Variables

> Linearity of Expectation $E[X+Y] = E[X] + E[Y]$

> $E[aX+b] = a\,E[X] + b$

> Two strategies to solve Expectation problems
  - Indicator variable (writing a r.v. as sum of multiple indicator variables)
  - Monte-Carlo chain method (writing a recursive eqn for expectation)

Joint PMF

$$P(x, y) = P(X = x \text{ and } Y = y)$$

$$\sum_{(x,y)} \sum P(x, y) = 1$$

Joint CDF

$$F(x, y) = \sum_{x_i \le x} \sum_{y_i \le y} P(X = x_i \text{ and } Y = y_i)$$

Marginal PMF

$$P_x(x) = \sum_j (x, y_i)$$

Independent rvs

If the joint PMF factors as the product of marginal PMFs, then the rvs are independent.

$$P(x_1, \dots, x_n) = P(x_1) \cdots P(x_n)$$

If X and Y are independent,

$$E[XY] = E[X] \cdot E[Y]$$

Expectation of functions of a rv

$$E[g(X)] = \sum_i g(x_i) \cdot P(x_i)$$

Expectation of functions of jointly distributed discrete rvs

$$E[g(X, Y)] = \sum_x \sum_y g(x, y) \cdot p(x, y)$$

Conditional probability distributions

$X \mid Y$ is also a rv for a given value of Y = y.

It takes the same values as rv X.

$$P_{X|Y}(x, y) = \frac{P(X = x \text{ and } Y = y)}{P(Y = y)} = \frac{P(X, Y)}{P(Y)}, \quad P(Y) > 0$$

Conditional expectation

$$E(X \mid \underbrace{F}_{\text{some event}}) = \sum_i x_i \cdot P(X = x_i \mid F) \qquad X \mid F: \ \Omega = \{x_1, x_2, \dots\} = \Omega_x$$

Theorem

$E[X] = \sum_i E[X \mid F_i] \cdot P[F_i]$ where $F_1, F_2, \dots, F_n$ are mutually exclusive and exhaustive events.

Ily. $E[X] = \sum_y E[X \mid Y = y] \cdot P(Y = y)$

E[X|Y] is a random variable and a function of rv Y.

$E[E[X|Y]] = E[X]$ ← Law of iterative expectations

Variance of DRV

$$Var(X) =$$

$Var(aX+b) = a^2 var(X)$

If X and Y are independent rvs,

$$Var(X+Y) = var(X) + var(Y)$$

Conditional variance

$$Var(X|Y=y) = E[(X - E[X|y])^2 | Y] = E[X^2|Y] - (E[X|y])^2$$

$$E[(X-\mu_{X|y})^2 | y] = \sum_i (x_i - \mu_{X|y})^2 \cdot P(X=x_i | Y=y)$$

E[X|Y] is a rv and a function on Y.
E[E[X|Y]] and Var(E[X|Y]) are well defined.

Var(X|Y) is a function on Y and a rv.
Y=y_i -> Var(X|Y=y_i) with prob P(Y=y_i)

E[Var(X|Y)] and Var(Var(X|Y)) are also defined.

Conditional variance/ law of total variance/ variance decomposition

$$Var(X) = E[var(X|Y)] + var(E[X|Y])$$

Covariance

$$cov(X,Y) = \sum_{x,y \in S} \sum (x-\mu_x)(y-\mu_y) \cdot P(x,y)$$

$$cov(X,Y) = E[(x-\mu_x)(Y-\mu_y)]$$
$$= E[XY] - E[X] \cdot E[Y]$$

If X and Y are independent, cov(X, Y) = 0.

$$cov(X,Y) = cov(Y,X)$$

$$cov(X,X) = var(X)$$

$$cov(aX, Y) = a\, cov(X,Y)$$

$$cov(X+Y, Z) = cov(X,Z) + cov(Y,Z)$$

$$cov(\sum X_i, \sum Y_j) = \sum_i \sum_j cov(X_i, Y_j)$$

$$cov(X+c, Y) = cov(X,Y)$$

$$var(X+Y) = var(X) + var(Y) + 2cov(X,Y)$$

$$var(\sum X_i) = \sum var(X_i) + 2\sum\sum_{i<j} cov(X_i, X_j)$$

Correlation

$$\rho_{X|Y} = \frac{cov(X,Y)}{\sigma_x \sigma_y}$$

Random Variable	Ω	PMF	Expectation	Variance
Discrete Uniform (a, b)	$x = a, a+1, \dots, b$	$P(X=x) = \frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2 - 1}{12}$
Bernoulli (p)	$x = 0, 1$	$P(X=1) = p$ $P(X=0) = 1-p = q$	p	$p(1-p) = pq$
Binomial (n, p) (# of successes in n independent Bernoulli trials)	$x = 0, 1, \dots, n$	$P(X=x) = {}_nC_x\, p^x (1-p)^{n-x}$	np	$np(1-p) = npq$
Poisson (λ) (binomial for large n and small p with lambda = np)	$x = 0, 1, \dots, \infty$	$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}$	λ	λ
Geometric (p) (# of Bernoulli trials to get a success)	$x = 1, 2, \dots, \infty$	$P(X=x) = (1-p)^{x-1} \cdot p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial (r, p) (# of Bernoulli trials to get r successes) (sum of r Geometric rvs)	$x = r, r+1, \dots, \infty$	$P(X=x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Hypergeometric (n, N, m) m white, N-m black → n w/o replacement X: # of white balls	$x = 0, 1, \dots, m$	$P(X=x) = \frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{m}}$	$np = n\frac{m}{N}$	$npq\left(1 - \frac{n-1}{N-1}\right)$ $= n\frac{m}{N}\left(1-\frac{m}{N}\right)\left(1-\frac{n-1}{N-1}\right)$